Chapter 19: Electric Potential & Electric Energy

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PHY201
Lecture 19

Part I

Electric Potential Energy

Outline of Part I: Electric Potential Energy

Electric Potential and Potential Difference

Outline of Part I: Electric Potential Energy

Electric Potential and Potential Difference

Electric Potential Energy

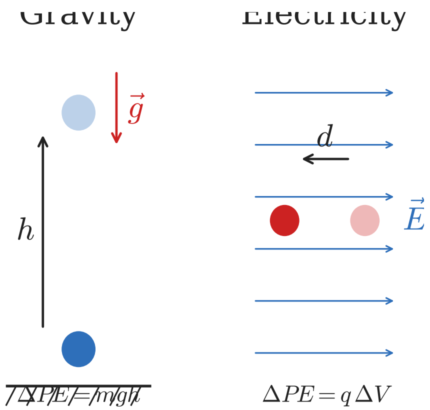
Recall: Gravitational PE

Moving a mass *against* gravity increases gravitational PE.

Moving a charge *against* an electric force increases **electric PE**.

Analogy

Gravity	**Electricity**
$\Delta PE_{\text{grav}} = mgh$	$\Delta PE_{\text{elec}} = q\Delta V$



Electric Potential V

Electric Potential — SI Unit: volt (V = J/C)

Electric potential V is the electric potential energy *per unit charge*:

$$V = \frac{PE}{q}$$

- $[V] = \mathrm{J/C} = \text{volt (V)}$
- V is a *scalar* — magnitude but no direction
- Reference point: $V = 0$ at infinity (or ground)

High $V \Rightarrow$ high potential energy per charge

Low $V \Rightarrow$ low potential energy per charge

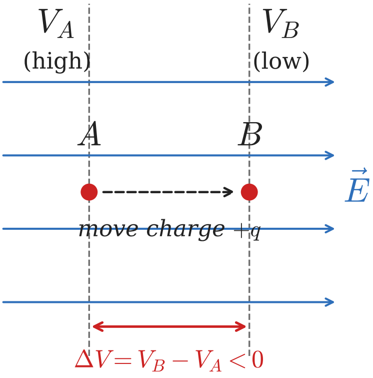
Potential Difference (Voltage)

ΔV — SI Unit: volt (V)

$$\Delta V = V_B - V_A = \frac{\Delta PE}{q}$$

$$\Delta PE = q\Delta V$$

- Voltage always measured *between two points*.
- "Voltage" is shorthand for potential difference.



Check Your Understanding

Which is more descriptive: "voltage" or "potential difference"?

Come to lecture to see the answer.

Check Your Understanding

If $\Delta V = 0$ between two points, does that mean zero work was done moving a charge between them?

Come to lecture to see the answer.

The Electron Volt (eV)

Electron Volt — 1 eV $= 1.60 \times 10^{-19}$ J

One **electron volt** is the energy gained by one electron moving through a potential difference of 1 V:

$$1\,\text{eV} = (1.60 \times 10^{-19}\,\text{C})(1\,\text{V}) = 1.60 \times 10^{-19}\,\text{J}$$

Atomic-scale energies in joules are inconveniently tiny.

1 eV is a natural unit for electrons, atoms, and nuclei.

An electron accelerated through 100 V gains

$100\,\text{eV} = 1.60 \times 10^{-17}\,\text{J}$.

Check Your Understanding

What is the difference between an *electron volt* and a *volt*?

Come to lecture to see the answer.

Example: Battery Energy — (cp2e_ch19_ex1)

A 12-V car battery moves 60,000 C of charge. How much energy is delivered? Compare with a motorcycle battery that moves 40,000 C through the same voltage.

$$\text{Car:} \quad \Delta PE = q\Delta V$$
$$= (60{,}000)(12)$$
$$\boxed{\Delta PE = 720{,}000\,\text{J}}$$

$$\text{Moto:} \quad \Delta PE = (40{,}000)(12)$$
$$\boxed{\Delta PE = 480{,}000\,\text{J}}$$

Same voltage, different charge moved $\Rightarrow$ different energy delivered.

A battery's voltage alone doesn't tell you its total stored energy — you also need to know how much charge it can move.

Check Your Understanding

A negative charge is released from rest in a uniform electric field. Does it move toward higher or lower electric potential?

Come to lecture to see the answer.

Part II

Electric Potential in Different Configurations

Outline of Part II: Electric Potential in Different Configurations

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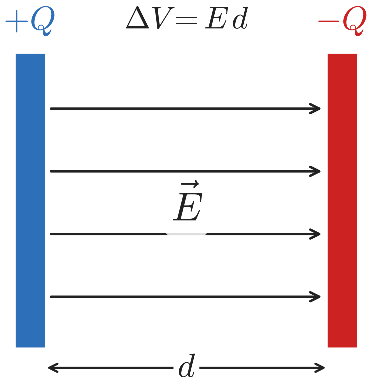
Potential Difference in a Uniform Field

Uniform Field Between Parallel Plates

$$\Delta V = Ed \qquad E = \frac{\Delta V}{d}$$

- $\vec{E}$ from high V to low V (+ plate to − plate).
- Stronger field ⇒ steeper potential drop.
- $[E] = \mathrm{V/m} = \mathrm{N/C}$ (both valid SI).



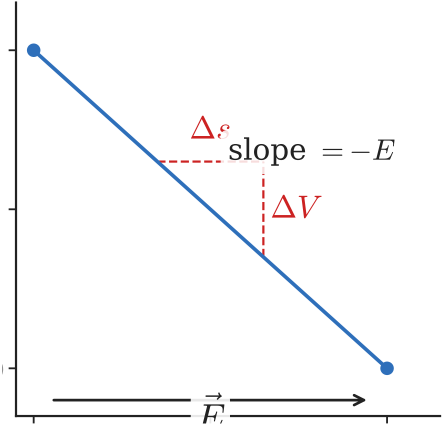
General Field–Potential Relationship

Non-Uniform Fields

Along displacement Δs:

$$E = -\frac{\Delta V}{\Delta s}$$

$\vec{E}$ points toward *decreasing* V (negative sign).



- Largest E where V changes most steeply.
- $E = 0$ where V is constant.
- Field lines point "downhill" in potential.

Check Your Understanding

If the electric potential is constant throughout a region, what is the electric field there?

Come to lecture to see the answer.

Check Your Understanding

A positive charge is placed between two parallel plates (+ plate on right, − plate on left). Toward which plate does it accelerate?

Come to lecture to see the answer.

Outline of Part II: Electric Potential in Different Configurations

Potential Due to a Point Charge

Electric Potential of a Point Charge — SI Unit: volt (V)

The electric potential at distance r from a point charge Q:

$$V = \frac{kQ}{r}$$

where $k = 8.99 \times 10^9\,\mathrm{N{\cdot}m^2/C^2}$.

- V is a *scalar* — add contributions from multiple charges algebraically.
- $V > 0$ for positive Q; $V < 0$ for negative Q.
- $V \to 0$ as $r \to \infty$ (reference at infinity).
- $V \propto 1/r$ (falls off slower than $E \propto 1/r^2$).

Equipotential surfaces are concentric spheres around a point charge.

Superposition of Potentials

Multiple Charges — Scalar Sum

Total potential is a *scalar* sum (no vector addition needed!):

$$V_{\text{total}} = \frac{kQ_1}{r_1} + \frac{kQ_2}{r_2} + \cdots$$

Much simpler than superposing $\vec{E}$ fields!

Potentials just add as signed numbers — no angle or component work needed.

$+Q$ and $-Q$ each at distance a from midpoint P:

$$V = \frac{kQ}{a} + \frac{k(-Q)}{a} = 0$$

(but $\vec{E} \neq 0$ there!)

Check Your Understanding

Two equal and opposite charges are equidistant from point P. What are V and $\vec{E}$ at P?

Come to lecture to see the answer.

Check Your Understanding

A point charge creates $V = 100\,\mathrm{V}$ and $E = 50\,\mathrm{N/C}$ at distance r. What are V and E at distance $2r$?

Come to lecture to see the answer.

Outline of Part II: Electric Potential in Different Configurations

Uniform Electric Field

Electric Potential of a Point Charge

Equipotential Lines

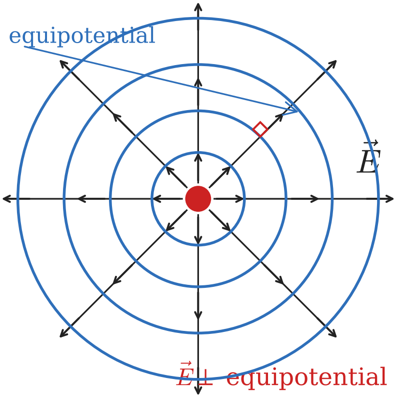
Equipotential Lines and Surfaces

Equipotential Lines

An **equipotential line** (or surface) connects all points with the *same* electric potential.

Key Properties

- Always **perpendicular** to electric field lines.
- No work done moving a charge *along* an equipotential ($\Delta V = 0$).
- Equipotential lines **cannot cross**.
- Conductors in electrostatic equilibrium are equipotential surfaces.



Check Your Understanding

Can two equipotential lines ever cross each other? Why or why not?

Come to lecture to see the answer.

Check Your Understanding

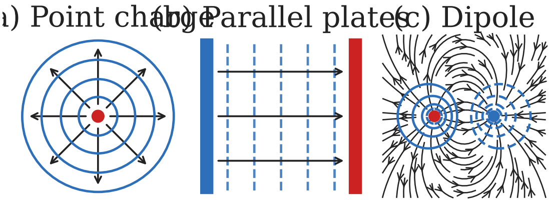
How much work is done by the electric force when a charge moves along an equipotential surface?

Come to lecture to see the answer.

Equipotentials Around Common Charge Distributions

- **Point charge:** concentric circles
- **Parallel plates:** parallel lines
- **Dipole:** complex curves

Field lines always *perpendicular* to equipotentials.



Part III

Capacitors

Outline of Part III: Capacitors

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Capacitors

Capacitance C — SI Unit: farad (F = C/V)

A **capacitor** stores electric charge on two conducting plates separated by an insulator. Its **capacitance** is:

$$C = \frac{Q}{V}$$

where Q = charge stored (C) and V = voltage across plates (V).

- C depends *only* on geometry and material — not on Q or V.
- Larger $C \Rightarrow$ more charge stored per volt.

$$C = \frac{Q}{V}$$

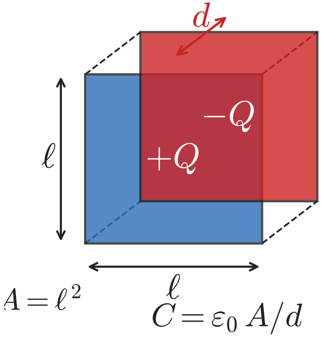
Typical values: pF to μF to mF

Capacitance of a Parallel-Plate Capacitor

Geometry Formula: $C = \varepsilon_0 A/d$

- $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F/m}$
- $A =$ plate area; $d =$ separation

- Larger $A \Rightarrow$ more overlap $\Rightarrow$ larger C.
- Smaller $d \Rightarrow$ stronger field $\Rightarrow$ larger C.



Check Your Understanding

Does capacitance depend on the voltage applied or the charge stored? Explain.

Come to lecture to see the answer.

Dielectrics

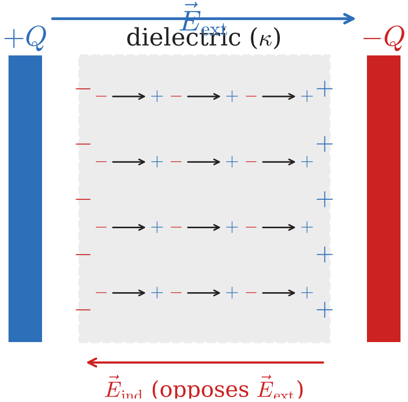
Dielectric Constant κ

Insulator between plates multiplies C by κ:

$$C = \kappa\,\varepsilon_0 \frac{A}{d}$$

Polarized molecules create opposing field, reducing E. For same Q, lower $E \Rightarrow$ lower V, so $C = Q/V$ increases.

$\kappa > 1$ always.



Check Your Understanding

A capacitor is connected to a battery and fully charged, then *disconnected*. The plates are then pulled farther apart. Does the charge on the plates increase, decrease, or stay the same?

Come to lecture to see the answer.

Check Your Understanding

A dielectric is inserted into an *isolated* (disconnected) capacitor. Does the voltage across it increase, decrease, or stay the same?

Come to lecture to see the answer.

Outline of Part III: Capacitors

Capacitors in Parallel

Parallel Combination

Same voltage across each capacitor. Total capacitance:

$$C_P = C_1 + C_2 + \cdots + C_n$$

- More plates in parallel ⇒ effectively larger plate area ⇒ larger C.
- Capacitors in parallel add like *resistors in series*.

$$C_P = \sum C_i$$

Parallel always gives $C_P >$ any individual C_i.

Capacitors in Series

Series Combination

Same charge on each capacitor. Total capacitance:

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

- Series increases effective plate separation $\Rightarrow$ smaller C.
- Capacitors in series add *reciprocally* like resistors in parallel.

$$\frac{1}{C_S} = \sum \frac{1}{C_i}$$

Series always gives $C_S <$ smallest C_i.

Check Your Understanding

Three identical capacitors, each C, are connected in *parallel*. What is the equivalent capacitance?

Come to lecture to see the answer.

Check Your Understanding

Three identical capacitors, each C, are connected in *series*. What is the equivalent capacitance?

Come to lecture to see the answer.

Example: Mixed Capacitor Network — (cp2e_ch19_ex2)

$V = 12\,\mathrm{V}$; $C_1 = 1\,\mathrm{F}$ in series with $(C_2 = 2\,\mathrm{F}$ parallel with $C_3 = 3\,\mathrm{F})$. Find the charge on C_3.

Step 1 — Combine $C_2 \parallel C_3$:

$$C_{23} = 2 + 3 = 5\,\mathrm{F}$$

Step 2 — Combine C_1 series C_{23}:

$$\frac{1}{C_{\mathrm{eq}}} = \frac{1}{1} + \frac{1}{5} = \frac{6}{5} \Rightarrow C_{\mathrm{eq}} = \frac{5}{6}\,\mathrm{F}$$

Step 3 — Total charge:

$$Q_{\mathrm{tot}} = C_{\mathrm{eq}}\, V = \tfrac{5}{6}(12) = 10\,\mathrm{C}$$

Step 4 — Voltage across C_{23}:

$V_1 = Q/C_1 = 10/1 = 10\,\mathrm{V}$

$V_{23} = 12 - 10 = 2\,\mathrm{V}$

Step 5 — Charge on C_3:

$$Q_3 = C_3\, V_{23}$$

$$\boxed{Q_3 = (3)(2) = 6\,\mathrm{C}}$$

Outline of Part III: Capacitors

Energy Stored in a Capacitor

Stored Energy — SI Unit: joule (J)

Three equivalent forms:

$$E_{\text{cap}} = \frac{QV}{2} = \frac{CV^2}{2} = \frac{Q^2}{2C}$$

Energy is stored in the **electric field** between the plates, not in the charges themselves.

Energy density: $u = \frac{1}{2}\varepsilon_0 E^2$

Defibrillator: stores 400 J at 10,000 V:

$$C = \frac{2E}{V^2} = \frac{2(400)}{(10^4)^2}$$

$$= 8.00\,\mu\text{F}$$

Check Your Understanding

For maximum energy storage, should capacitors be connected in series or in parallel? Explain.

Come to lecture to see the answer.

Check Your Understanding

The voltage across a capacitor is doubled. By what factor does the stored energy change?

Come to lecture to see the answer.

Check Your Understanding

An isolated capacitor has its plates pulled farther apart (increasing d). Does the stored energy increase, decrease, or stay the same?

Come to lecture to see the answer.

Check Your Understanding

A dielectric is inserted into an isolated capacitor. Does the stored energy increase, decrease, or stay the same?

Come to lecture to see the answer.

Check Your Understanding

A dielectric is inserted into a capacitor that *remains connected to a battery.* Does the stored energy increase, decrease, or stay the same?

Come to lecture to see the answer.

Chapter Summary

Equation	Variables	Conditions
$V = PE/q$	V: potential (V); PE: energy (J)	definition
$\Delta PE = q\Delta V$	q: charge (C)	work by electric force
$1\,\text{eV} = 1.60 \times 10^{-19}\,\text{J}$	electron volt unit	atomic-scale energy
$\Delta V = Ed$	E: uniform field; d: separation	parallel plates
$E = -\Delta V/\Delta s$	field–potential relation	general
$V = kQ/r$	$k = 8.99 \times 10^9$; r: distance	point charge
$C = Q/V = \varepsilon_0 A/d$	A: area; d: separation	parallel-plate capacitor
$C = \kappa\,\varepsilon_0 A/d$	κ: dielectric constant	with dielectric
$C_P = \sum C_i$	parallel combination	same V across each
$1/C_S = \sum 1/C_i$	series combination	same Q on each
$E_{\text{cap}} = \frac{1}{2}CV^2 = Q^2/(2C)$	stored energy (J)	capacitor